

The algebraic axioms, and what defines them in the Frobenius calculus

Freyd, Categories, *Algebraic* §11.4-11.6 against functorialSemanticsForRelationalTheories.pdf [Nc 4]
Lean: Freyd/S2_10.lean[Algebra],diag/CB.lean[CartBicat],diag/CB_Algebra.lean[algebraOfCartBicat]

Where the pictures come from. Every picture of a theorem is omitted by ./scripts/diag-export from that theorem's Lean type. Nothing about a statement is redrawn here. The two figures marked nonredrawn are the exception: they copy the paper's definition pictures, and definition has no statement to export.

Equation numbering. The numbers below are the 2017 preprint's, which is what diag/CB.lean's field names encode (ineq_37 ... ineq_40). The repo's copy of the paper is now the published version – Poly J Math, 2 (6), 20 Oct 2025, doi:10.6970/polyjmath.2.6 – which renumbers. Translation, published as – preprints (39)–(37), (35)–(40), (39)–(40), (39)–(41), (39)–(42), (40)–(43), (33)–(35), and Example 2.36(1)–(13) (11)–(11), (12); the preprint's (36) is unnumbered there. Pages: Def. 4.1 at 20–21 (was 17), the T notation at 21 (was 19), Lemma 4.2 at 21–22 (was 19). Renumbering the Lean fields is a separate decision and has not been made.

Freyd's allegory takes reciprocation * and intersection n as primitive operations and lists eight equational axioms on them. A cartesian bicategory of relations takes them: every object carries one special Frobenius structure Δ, ∇, τ , the hom-order $\leq$ is primitive, and both * and n are **defined** from the four dots. Every one of Freyd's eight axioms is then a theorem. The table below is that dictionary, one row per axiom.

Reading the pictures: left to right is composition – the book's diagram order, R $\circ$ S means first R then S – vertical stacking is $\otimes$, a bare wire is an identity, a white box is a relation, and a box that is mirrored and tinted is its converse – the mirroring is the paper's own notation, the tint is this note's, because at a glance a chamfer is easy to miss. The label inside is **not** mirrored: the paper can reverse a single letter and still be read, a whole term like R $\cup$ S cannot. All four dots are solid, as Def. 4.1 draws them. The exporter draws a meet or a converse as a labelled block, so a composite under Δ appears as a single box reading R ; S ; a $\otimes$ draws as two stacked runs; and any subterm with no shape in this calculus – an associator, say – degrades to an opaque box. The term columns write the converse R^* , Freyd's symbol, where the paper writes R†: these terms are read against the book throughout, and one symbol per idea beats asking the reader to match two. (The Lean doctstrings keep †, deliberately – they cite the paper, whose own (–)† is the \$7 colour swap and is not order-reversing.)

In a chain table the symbol at the head of a proof is the relation of that step was proved at: = for a reshaping, $\leq$ for a genuine inequality. Counting them the point of leading each arrow at all – a chain of two $\leq$ between equalities says the theorem is one law and a lot of bookkeeping.

1. Freyd's primitives are the Frobenius calculus's definitions

Freyd's primitive, and what defines it here	picture
reciprocation R^* – §2.11, primitive. Here conv $R := \text{bend } ((R \otimes 1) ; \text{cap}) (\text{diag/CB.lean:723})$, not a generator. Input arrives bottom left and runs under R; the cup ∇ ; Δ at the top left creates a pair, one strand feeding R and the other leaving as the output at the top right: the cup ∇ ; Δ at the bottom right annihilates R's output against the input. It is the cup and the cap that reverse the arrow – a wire merely bumped up over the box and back down is an isotopy of R itself.	
intersection $R \cap S$ – §2.11, primitive. Here the convolution meet $R \cap S := \Delta ; (R \otimes S) ; \nabla (\text{diag/CB.lean:24})$. Copy the input, run both, merge; the merge forces the two results to agree.	
containment $R \subseteq S$ – §2.11, defined as $R \cap S = R$. Here the poset-enrichment $\leq$ is primitive and n is derived, so the two directions are swapped. meet_eq_left_of_le (diag/CB.lean:196) is the bridge.	no picture – this is the 2-cell itself
maximal morphism $\dashv$ – §2.15, and Freyd needs a unitary allegory for it. Here top $a \cdot b := ! ; \nabla$ (diag/CB.Derived.lean:28) is a free in Def. 4.1. Freyd.Diag.lean_top	

2. The eight axioms, and what proves each

Freyd's axiom, and what supplies it	picture, from the Lean statement
$(R^*)^* = R$ – §2.11 recip_recip: here the snake, from (41). Freyd.Diag.CartBicat.conv_conv. Bending down and back up straightens out, so bending twice is the identity.	
$(R \cap S)^* = S^* \circ R^*$ – §2.11 recip_comp; here mirroring the picture. Freyd.Diag.CartBicat.conv_comp. Reflecting a chain left to right reverses it; each box comes back flipped.	
$(R \cap S)^* = R^* \cap S^*$ – §2.11 recip_inter; the same mirroring. Freyd.Diag.conv_inter. Δ and ∇ are each other's mirror image, so reflecting the convolution leaves its shape alone and only turns the boxes round.	
$R \cap R = R$ – §2.11 inter_idem; here (42) and the special law. Freyd.Diag.meet_idem. Copy, run R twice, merge – (42) pushes R back through the copy and Δ ; ∇ = 1 closes the bubble.	
$R \cap S = S \cap R$ – §2.11 inter_comm; here commutativity (9) and commutativity (46). Freyd.Diag.meet_comm. The two strands of Δ are interchangeable, and so are those of ∇ .	
$R \cap (S \cap T) = (R \cap S) \cap T$ – §2.11 inter_assoc; here coassociativity (8) and associativity (5). Freyd.Diag.meet_assoc. Either bracketing builds the same three-way copy tree, and dually the same merge tree.	
semi-distributivity $R (S \cap T) \subseteq R S \cap R T$ – §2.11 semidistrib, §2.136; here (42), the lax comonoid inequation. Freyd.Diag.semistrib_of_lax. Running R and then copying is below copying and then running R on each strand.	
modular identity $R S \cap T \subseteq (R \cap T) S^*$ – §2.12 modular; the axiom Freyd adjoins: "we adjoin one more containment, the law of modularity". Here (41) and (42), derived. Freyd.Diag.modular_of_frobenius. Strip the shared prefix Δ ; $(R \otimes T)$ off both sides and what is left is nabla_slide_conv (diag/CB.lean:556) – a box slides through a merge at the cost of its converse on the other strand.	

3. The other direction: what each law of Def. 4.1 says in Freyd's language

picture, from the Lean statement	Def. 4.1	Freyd
	(37) $\nabla ; \Delta \leq 1$ half of $\Delta \dashv \nabla$	$\Delta^* \subseteq 1 - \Delta$ is simple §2.13. Not used anywhere in the bridge; see below.
	(38) $1 \subseteq \Delta ; \nabla$ the other half of $\Delta \dashv \nabla$	$1 \subseteq \Delta - \Delta^*$ is entire §2.13, with (37). Δ is a map.
	(39) $! ; \nabla \leq 1$ half of $! \dashv \nabla$	$!^* \subseteq 1 - !$ is simple §2.13. Also unused in the bridge.
	(40) $! \nabla \leq ?$ the other half of $! \dashv \nabla$	$1 \subseteq !^* - !$ is entire §2.13, with (39). $!$ is a map. This is what makes $T = ! ; \nabla$ contain every arrow.
	(41) the Frobenius law an axiom. $! \otimes \Delta$ and $\nabla \otimes !$ draw as stacked runs; only the associator α^{-1} stays an opaque box, having no shape in this calculus.	the modular identity §2.112 – and reciprocation §2.11 itself, since the cup and cap only satisfy the snake because of (41).
	the special law not as axiom: derived from (38) and (40). Freyd.Diag.CartBicat.special	$\Delta \Delta^* = 1 - \Delta$ is injective, so by §2.141 at $f = g = \Delta$ is a monic map. In Rel(Set): copying is total and loses nothing.
	(42) every arrow is a lax comonoid homomorphism	§2.136's $R (S \cap T) \subseteq R S \cap R T$, read at $S := n_1^*, T := n_2^*$ – Freyd's semi-distributivity, product-free.
	(43) every arrow is lax for the counit	§2.152's step $p_{\alpha S}$ is maximal in (α, λ) , hence $R \cdot p_{\alpha S} \subseteq p_{\alpha^*} -$ that is, $R \subseteq T$.

4. Unitary and pre-tabular – the two conditions the converse needs

Freyd's §2.11 axioms are all a cartesian bicategory has to give back. Going the other way needs two further conditions, and they are exactly the two the repo has not formalised. Both definitions below are quoted from Freyd/S2_10.lean, which follows the book.

condition	Freyd	what the Frobenius side already has
unitary §2.13	T is a partial unit when $! \vdash T$ is the largest endomorphism on it. PartialUnit $T := Y \circ R : T \rightarrow T$, $R \subseteq T \vdash T$ (§399); it is a unit when moreover every object carries an entire arrow to it, $! \vdash \text{Unit}$ $T := \text{PartialUnit } T \wedge \forall a, \exists R : a \rightarrow T$, Entire R (§545). A UnitaryAllegory is an allegory with one (§548). The point of a unit is that it makes $T_{(a,b)}$ exist: $T = p_{\alpha} p_{\beta^*}$ through the unit, which is where §2.152's $R \cdot p_{\beta} \subseteq p_{\alpha^*}$ – the repo's (43) – comes from.	already there, for free. The monoidal unit I is the unit object, $! \otimes \Delta = 1$ is the entire arrow every object carries to $!$, (40) says so – and $T = ! ; \nabla$ is maximal by Freyd's §2.141, le_top. So every cartesian bicategory of relations is unitary, with unbiased assumed . Not recorded in Lean: allegoryOfCartBicat builds a plain Allegory, and no UnitaryAllegory instance is derived from CartBicat.
pre-tabular §2.165	f, g tabulate R when both are maps, $R = f^* \circ g$, and $f \circ f^* \circ g \circ g^* = 1$ (Tabulates, §502); R is tabular when some pair tabulates it (§506). An allegory is PreTabularAllegory when every morphism is contained in a tabular one (§556). §2.165: "If Δ is unitary then pre-tabularity is equivalent to the existence of tabulations for maximal morphisms".	this is what $\otimes$ is. Given unitarity, the condition reduces to tabulating each $T_{(a,b)}$, and a tabulation of $T_{(a,b)}$ is precisely an object $a \otimes b$ with its two projections – the monoidal product Def. 4.1 assumes. That is the whole content of the swap: Freyd derives the product from tabulations, §4.1 posits it. Not built: RelBuilding $\otimes$ from tabulations and products in Map(A) is CartBicatOfUnitaryPreTabular; diag/PLAN, and holds it back for want of a consumer.

Together they are the whole gap: cartesian bicategory of relations $\equiv$ unitary pre-tabular allegory (DiagrammaticAlgebraOfFirstOrderLogic §10, after Carboni–Walters). One half of that equivalence is allegoryOfCartBicat and a theorem in this repo; the other half is the two rows above.

5. Union and residual – what phases 8 and 9 add

u and / are past where Freyd's §2.11 list stops, and past what Def. 4.1 can build: they need a **bi**product on top of the Frobenius structure (diag/Tape.lean, after TapeDiagrams.pdf) and a **second composition** with linear adjoints (diag/F0.lean, after DiagrammaticAlgebraOfFirstOrderLogic.pdf). Both are built, so both draw.

The two notations say which layer a picture is in. A union is Δ : the tape: the rounded wrapper is the second monoidal product, and its $\leq$ / $\geq$ open and close a branch; a particle takes exactly one of – the same fork-run-join shape as a meet, on a different product. A residual is a **boxed** box, because it is not a composite of the generators at all: $R / S := R ; S^*$ lives in the second composition, and a solid box would claim it was drawable in the first.

statement	picture, from the Lean statement
$R \subseteq R \cup S$ – §2.2 in Freyd; here Freyd.Diag.BiProd.lean_union_left. union $R \cup S := (R, S) ; [1, 1] (\text{diag/Tape.lean:84})$; offer both branches, then forget which was taken.	
$R \cup S = S \cup R$ – Freyd.Diag.BiProd.union_comm. The two branches of a tape are interchangeable, exactly as the two strands of Δ are.	
$\text{bot } a \cdot b := \text{toZero } a ; (\text{toZero } b)^*$ (diag/Tape.lean:87); routed through the zero object, the exporter has no shape for it, so it prints as a box.	
$R (S \cup T) = R S \cup R T$ – Freyd.Diag.BiProd.comp_union; composition distributes over union, which n does not. Freyd §2.2's distributivity, the reason a distributive allegory is more than an allegory.	
$(R \cup S)^* = R^* \cup S^*$ – Freyd.Diag.BiProd.conv_union. The converse is a 2-functor for the tape layer too.	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ – Freyd.Diag.FbCnMeetUnionDistrib. the law that makes distributiveAllegoryOfFbCb a distributive allegory (§2.2). One picture with both layers in it: meets inside a tape on the left, tapes around meets on the right.	
$(R / S) S \subseteq R$ – Freyd.Diag.ClosedLinearBicat.residual_comp_le; the residual is a lower bound of the arrows it is the greatest of. Freyd's §2.31 division R / S is the largest T with $T S \subseteq R$, and le_residual_iff (diag/F0.lean:170) is that Galois statement in full. The dashed frame marks the box as satisfying the second composition.	

6. Lemma 4.2 (ii) – the proof, not just the statement

diag-export – pre-prove a proof. A calc chain elaborates to nested trans, trans, so its intermediate terms are all recoverable from the proof term: flatten the spine, infer the type of each line, draw one canvas per step. Nothing is replayed and nothing is retold – the nine rows below are the chain the Lean kernel accepted for Freyd.Diag.CartBicat.conv_conv, which is Lemma 4.2 (ii). (R ; S)^* = S^* \circ R^*.

6.1. First: why these pictures are not pictures of a converse

They are not, and that is the thing to get straight before reading them. The converse is drawn with two bends – a cup on the way in and a cap on the way out, that pair of dots, ∇ ; Δ and ∇ ; Δ – which is the left-hand figure below. The proof never draws that. conv_unfold says an arrow is the converse of R as soon as it satisfies one equation between UNBENT arrows $a \otimes b \vdash !$: two strands in the on left, one cap on the right. Bending is a bijection (bend_unbend_unbend_eq), so setting the unbent equation satisfies the bent one, and the cup never has to be drawn at all.

That is why every cup below has to be drawn as a cap and no cup, and why an arrow that "is a converse" shows up not as a bent box but as a box sitting on the bottom strand, mirrored.

the converse, as the paper draws it	the one rule the proof uses, twice
R^* is R with both wires bent round: in at the bottom left, cap at the top left, ∇ at the bottom right, then Δ at the top right, as each bend, as functorialSemanticsForRelationalTheories.pdf draws it – a cup is ∇ , a cap is Δ then $!$. Transcribed by hand; a definition has no statement to export.	conv_slide (diag/CB.lean:449): a box on the BOTTOM strand facing a cap is the same as that box on the TOP strand, upright. Sliding round the bend is what turns R^* into R. This is the entire content Lemma 4.2 (ii). The other seven steps are bookkeeping.

6.2. the chain

One row per term, not per step: the right-hand side of every step is the left-hand side of the next, so drawing steps would draw every picture twice. The first column is the term as the Lean has it, fully bracketed, with $\otimes$ binding tighter than $;$ – it comes off the same walk as the picture, so the two cannot drift apart. Watch which strand each box is on: bottom means mirrored, top means upright.

Lemma 4.2 (ii), (R ; S)^* = S^* ; R^*	picture	the rule that reaches it
term $(R ; S)^* ; R^*$ picture 		the rule that reaches it about S†. Both conv_uniq and conv_start: $R^* \vdash R^*$ both boxes on the bottom strand, mirrored, so that is what "these are converses" looks like once the wires are unbent.
$= 1 \otimes S^* ; (1 \otimes R^* ; \text{cap})$ $= 1 \otimes S^* ; (R \otimes 1) ; \text{cap}$		1. $\otimes$ is a functor, so doing two things to one strand in sequence is the sequence of doing them. Same picture.
$= (R \otimes 1) ; \text{cap}$		2. conv_slide, R† is the box nearest the cap, so it slides round the bend: it leaves the bottom strand, arrives on the top, and comes back upright as R. The first of the two steps that prove anything.
$= (1 \otimes S^* ; R \otimes 1) ; \text{cap}$		3. simp only [Cat.assoc]. Re-bracketing – subtle in the term column, invisible in the picture. Same picture.
$= R \otimes S^* ; \text{cap}$		4. tenshom, split† – interchange. R is on the top strand and S† on the bottom, so running them side by side are the same thing. Shows as a re-spacing only because the exporter lays a diagram out left to right.
$= (R \otimes 1 ; 1 \otimes S^*) ; \text{cap}$		5. tenshom, split† – the same law, back the other way. Split them apart again. R first, so that S† is now the box facing the cap. Which is what the next real step needs.
$= R \otimes 1 ; (1 \otimes S^* ; \text{cap})$		6. simp only [Cat.assoc]. Re-bracketing. Same picture.
$= R \otimes 1 ; (S \otimes 1 ; \text{cap})$		7. conv_slide, on S this time. S† slides round the bend and arrives upright on the top strand. Both boxes are now on top, both upright, in the order R then S. The second and last real step.
$= (R \otimes 1 ; S \otimes 1) ; \text{cap}$		8. simp only [Cat.assoc]. Re-bracketing. Same picture.
$= (R ; S) \otimes 1 ; \text{cap}$		9. tenshom, comp† – interchange one more, the top-strand boxes merge into one. This is R ; S unbent, so conv_uniq ends up $S^* \circ R^* = (R ; S)^*$, Done.

What the chain measures. Read down the pictures and most consecutive pairs are the same drawing – steps 1, 3, 6, 8 and 9 change nothing at all, and step 4 changes only spacing – while the term column beside them changes at every row. So these are associativity, the unit laws and interchange – precisely the laws a string diagram has built into its geometry rather than as rewrites. Seven of these nine Lean steps are bureaucracy the picture language does not charge for, and the theorem is two applications of conv_slide.

How this differs from the paper's proof. The paper proves (ii) by inserting a snake in the middle of ∇ and (iv) and reading the two lines off as S† and R† – three pictures, all bent. The Lean never bends. Same theorem, different route, and the export shows the route the kernel accepted. Drawn by hand it would have shown the paper's three pictures and quietly implied those were what the repo had checked.

7. The modular law – proved, drawn

Freyd adjoins the modular identity: an allegory has no structure to derive it from, so it goes in as an axiom. Here it is a theorem, and this is its proof – Freyd.Diag.modular_of_frobenius (diag/CB_Algebra.lean:48), four terms, of which exactly one step is an inequality.

the modular law, $R S \cap T \subseteq (R \cap T) S^*$	picture	the rule that reaches it
term $(R ; S) \cap T$		the start: $R S \cap T$. A meet, so a copy and a merge with R Δ above and T below.
$= (\Delta ; R \otimes T) ; (S \otimes 1) ; \nabla$		Factor out the shared prefix. $(R \cap T) S^* = (R \otimes T) ; (S \otimes 1) ; \nabla$, so everything before the S is common to both sides and can be set aside. An equality, and drawn as one.
$\leq (\Delta ; R \otimes T) ; (1 \otimes S^*) ; (\Delta ; R \otimes T) ; (1 \otimes S^*) ; \nabla$		nabla_slide_conv , under the shared prefix. S slides through the merge and reappears as S† on the other strand. The only inequality in the proof – everything else is reshaping.
$= R \cap (T ; S^*) ; S$		Fold the prefix back in. $R \otimes (T ; S^*) = (R \otimes T) ; (1 \otimes S^*)$, and the copy-meet pair reads as a meet again. $(R \cap T ; S^*) S$, which is the goal.

7.1. Inside the middle step

nabla_slide_conv (diag/CB.lean:556) is where the mathematics is, and it mentions only S – R and T sit untouched in the prefix throughout. Read on its own it says one arrow slides through a merge:

term	picture	the rule that reaches it
$S \otimes 1 ; \nabla$		the start: $(S \otimes 1) \vdash \nabla$.
$= (S ; \Delta) \otimes 1 ; (\alpha ; (1 \otimes \text{cap}) ; \rho)$		nabla_of_cap, backwards. The merge is rebuilt from a copy and a cap, which puts the term into the shape $(S ; \Delta) \otimes 1$, that is (42) applies.
$\leq (\Delta ; S \otimes S) \otimes 1 ; (\alpha ; (1 \otimes \text{cap}) ; \rho)$		becomes Δ ; $(S \otimes S)$: the box is copied. The whole of the modular law, in one step.
$= 1 \otimes S^* ; (\nabla ; S)$		cap_tens, nabla back and conv_slide. Reshaping back, and the surviving duplicate slides round the cap to become S† – the same rule Lemma 4.2 (ii) is made of.

What the associators cost. α and ρ have no shape in this calculus and print as opaque boxes, so the two middle terms of the inner chain are illegible than its ends. That is the honest price of drawing what the Lean says: the reshaping steps really are that fiddly, and a hand-drawn version would have hidden them by working up to coherence.

8. Maps: SV, TOT, INJ, SUR, and the two inequations that always hold

Def. 4.1 clause 4 says every arrow is lax for Δ and for $!$ – in every (42) and (43). Applying Lemma 4.2 (ii) and (iv) to them gives (44) and (45), the same statements about ∇ and $?$. Those four hold for equality and equality. Their four REVERSES do not, and each one failing or holding is a property of the arrow: that is (SV), (TOT), (INJ), (SUR).

holds for every arrow	holds iff the arrow is ...
copy Δ $(43) R ; \Delta \leq \Delta ; (R \otimes R)$	(SV) – single valued $(\text{copy}) \vdash R^*$
discard $!$ $(43) R ; ! \leq !$	(TOT) – total $(\text{discard}) \vdash R$
merge ∇ $(44) \nabla ; R \leq (R \otimes R) ; \nabla$ – the converse of (42), not in the Lean; see below.	(INJ) – injective $(\text{merge}) \vdash R$
unit $?$ $(45) ? ; R \leq ?$ – the converse of (43). Not in the Lean; see below.	(SUR) – surjective $(\text{unit}) \vdash R$

The paper gives the same four properties a second time, as ADJOINT conditions – (46)–(49) – and Lemma 4.4 says the two presentations agree. diag/CB.Derived.lean takes the adjoint forms as the definitions, because they are literally Freyd's Simple and Entire (§2.13), so a diag map and a book map are one predicate with no translation layer.

property	adjoint form, the definition	Freyd
single valued Freyd.Diag.SingleValued		Simple $R := R^* \circ R \subseteq 1$ §2.13
total Freyd.Diag.Total		Entire $R := 1 \subseteq R \circ R^*$ §2.13, with single valued. R is a map.
injective Freyd.Diag.Injective		$R \circ R^* \subseteq 1$ §2.141 at $f = g = R$: a monic pair, so a monic map.
surjective Freyd.Diag.Surjective		§2.142: "g is a cover iff $1 \subseteq g^* \circ g$ ", that is, iff g^* is entire.

Why (44) and (45) have no picture. Both are converse-images, so drawing them from a Lean statement needs that statement to exist, and it does not. Getting it needs $R \vdash S^*$ – so "f is a map" and "f has a right adjoint, namely f†" are the same statement. That is Lemma 4.8 (Freyd.Diag.Lemma_4_8), which turns it round: any right adjoint of a map must be its converse, by the lemma which states it. (The paper's converse half – an arrow with a right adjoint is a map – is the diagram chase that needs Lemma 4.4, and is not proved here for the reason given above.)

9. Entrenchment of an intersection

Two definitions from Freyd, both used below. The domain of R is the part of the identity where R is defined, Dom $R = 1 \cap R \circ R^*$ (§2.122). R is entire iff it is defined everywhere, Dom $R = 1$, equivalently $1 \subseteq R \circ R^*$ (§2.13) – which is Total above, the picture opposite (47). The question this section answers is when an intersection is entire: Freyd's §2.124, "a lemma we will use repeatedly", and B&M 4.18.

term	picture	the rule that reaches it
1		the start: 1.
$\leq R \cap S ; (R \cap S)^*$		the hypothesis: $R \cap S$ entire. By the definition of entire, $1 \subseteq S \circ P^*$ at $P = R \cap S$.
$\leq R ; S^*$		monotonicity, twice: $R \cap S \leq R$ and $R \cap S \leq S$ under $?$. No modular law.

$\leq R \cap S ; (R \cap S)^*$		the hypothesis: $R \cap S$ entire. By the definition of entire, $1 \leq P \circ P^*$ if $P = R \cap S$.
$\leq R ; S^*$		monotonicity, twice: $R \cap S \leq R$, and $R \cap S \leq S$ under * , No modular law.

$\leftarrow$ given $\leq$		show $R \cap S$ entire,	$\leq$	
term		picture		